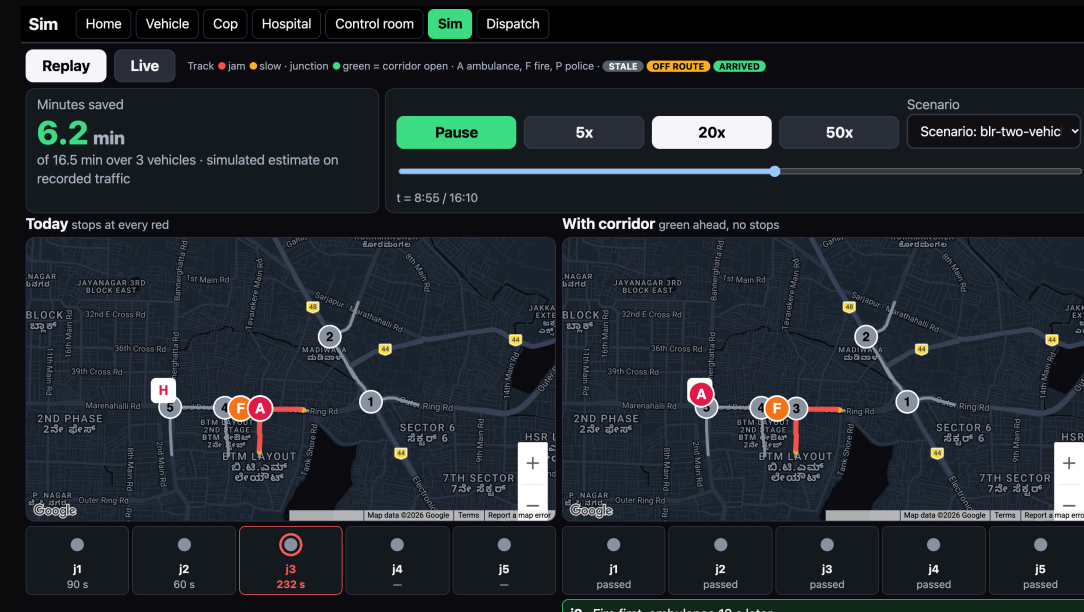


Emergency Green Corridor

Every red light costs a life.

Tell the cop at the next junction, early enough, that an ambulance or fire engine is coming.

Built by Nandish · Live at green-corridor-2026.web.app



The problem, seen daily on Hosur Road

7–10%

survival lost per minute in cardiac arrest without defibrillation

2–3 km

covered in 30–50 min by Bengaluru ambulances at peak hours

60%+

of Karnataka 108 cardiac, stroke and respiratory calls missed the 10-min target

The officer at the junction has no idea they are coming. Full signal preemption needs city controller integration: years away.

Sources: American Heart Association, CPR Facts and Stats (cpr.heart.org) and AHA, Circulation, 2017 · The Siasat Daily, 28 Jun 2026 (Karnataka Health Minister announcement) · CAG audit of Karnataka 108 EMS, 2014–19, reported by The News Minute, Dec 2020 (state-wide, not Bengaluru only)

The 20-second demo

Today: stops at every red.

With corridor: cop warned early, queue cleared.

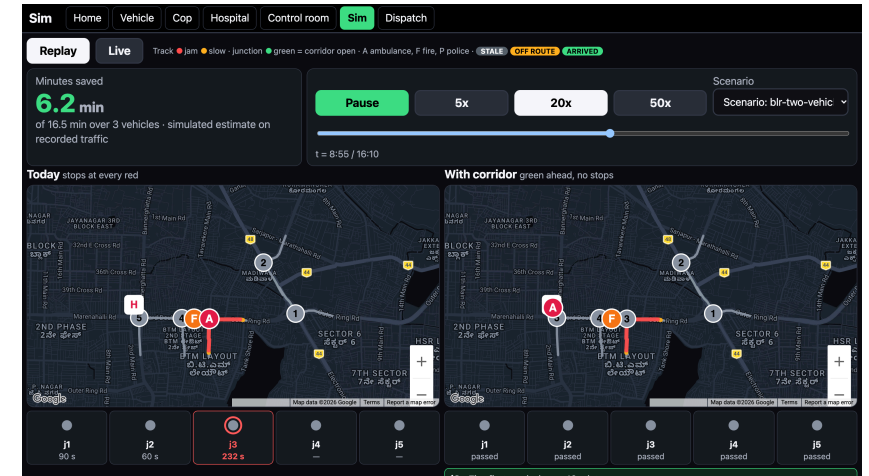
Silk Board → Jayadeva Hospital, the same scripted scenario in both lanes: GPS ticks generated along the real corridor roads, with hand-authored traffic spans (a 500 m queue at junction 3, 100 m at junction 4). The 3 vehicles run as one scenario: a critical ambulance, an ambulance following it as a platoon, and a fire engine.

≈ **16.5 min**

saved across 3 vehicles

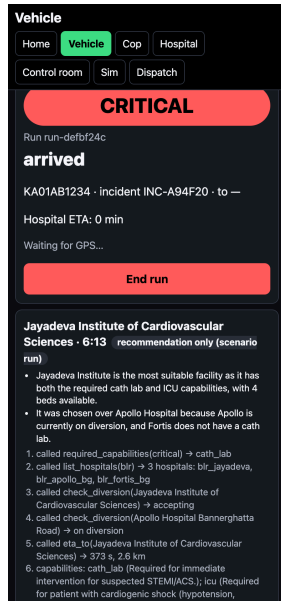
3

critical ambulance, platoon
ambulance, fire engine

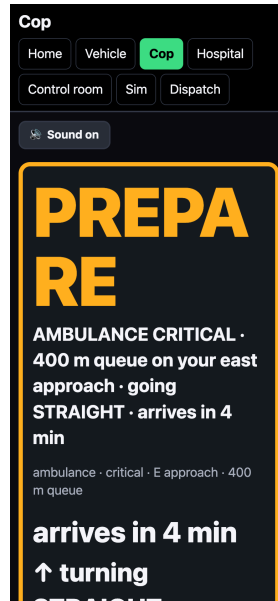


Simulated baseline on a scripted scenario, not a field measurement: per junction passed, cycle/4 + queue/2 m/s. Figure from the scenario replay (`blr-two-vehicles`). Live runs use live Routes traffic; scenario runs use the scripted spans.

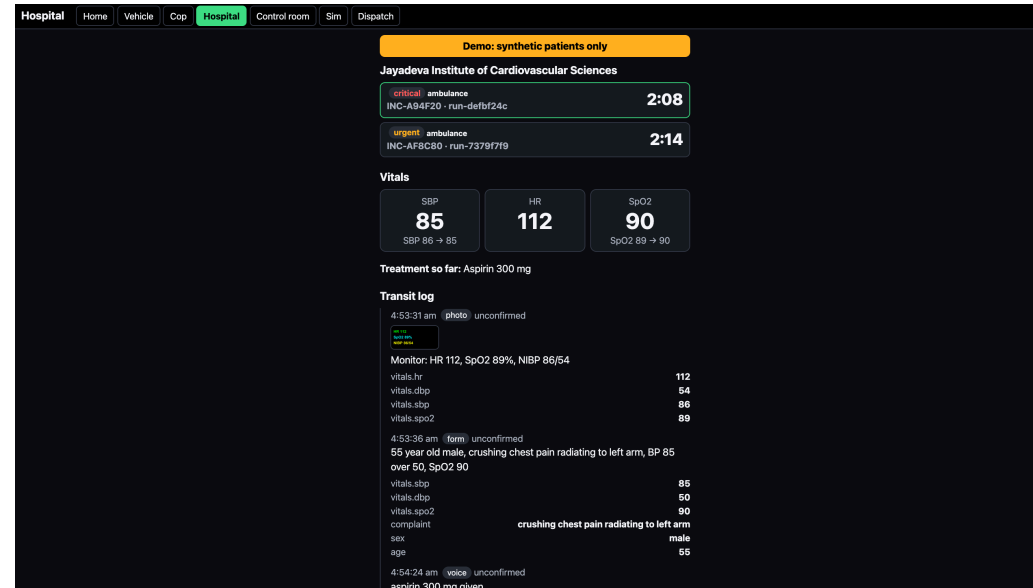
What it does: six screens



/vehicle
voice, tier tap



/cop spoken
alert, one ACK



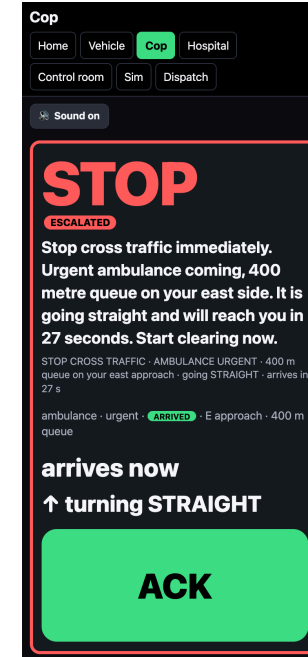
/hospital live log, vitals, treatment so far (the ATMIST brief lands at ETA-5)

Also: /dispatch incident IDs · /control all runs and escalations · /sim with-vs-without replay.

How a cop gets warned: the lead-time engine

Live runs: Routes traffic spans (scenario runs: scripted spans) → queue metres → clearance time → two-stage alert.

```
jam_m    = Σ JAM m + 0.5 × Σ SLOW m
clear_s  = 20 + jam_m / 2.0
eta_s    = 0.5 × routes_eta
          + 0.5 × distance / speed_60s
PREPARE          eta_s <= clear_s + 15
STOP CROSS TRAFFIC eta_s <= 30
```



No ACK in 20 s raises an escalation flag and an audit entry.

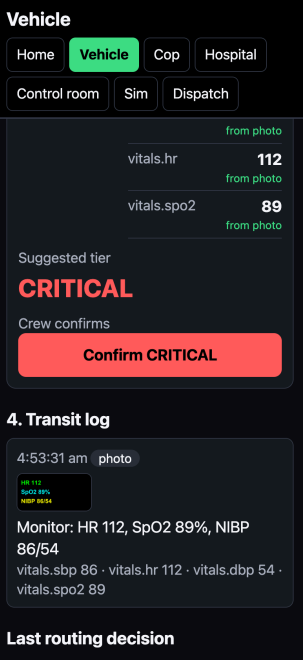
Rules decide, Gemini explains

Step	Who	What
1. Extract	Gemini	speech or monitor photo → fixed JSON fields, never a score
2. Tier	<code>acuity.py</code>	deterministic lookup → critical / urgent / stable; crew confirms with one tap
3. Order	<code>priority.py</code>	sort by (tier, ETA): a sequence, not a hold
4. Explain	Gemini	rewords the rule-built sentence for the cop ("Fire engine first, ambulance 12 s later"); validated, template if it fails

A language model must not decide who lives. Rules are auditable, testable and carry self-checks.

Where Gemini is load-bearing

Job	What
Voice and photo extraction	speech, typed text or a monitor photo into a fixed schema, with no tools.
ATMIST handover	the hospital brief and prep checklist, generated at ETA minus 5 minutes.
Alert phrasing	the short spoken line each junction cop hears.
Sequencing sentence	Gemini rewords a rule-built template into one line; the result is validated and the template is used if it fails.
Cop voice notes to rule actions	a spoken report from the junction fills a fixed schema, and plain rules act on it.
After-action report	a plain summary of each finished run, with the timeline built in code.
Hospital routing agent	built on Agent Development Kit, with four tools and a code guard that checks its choice before it is applied.



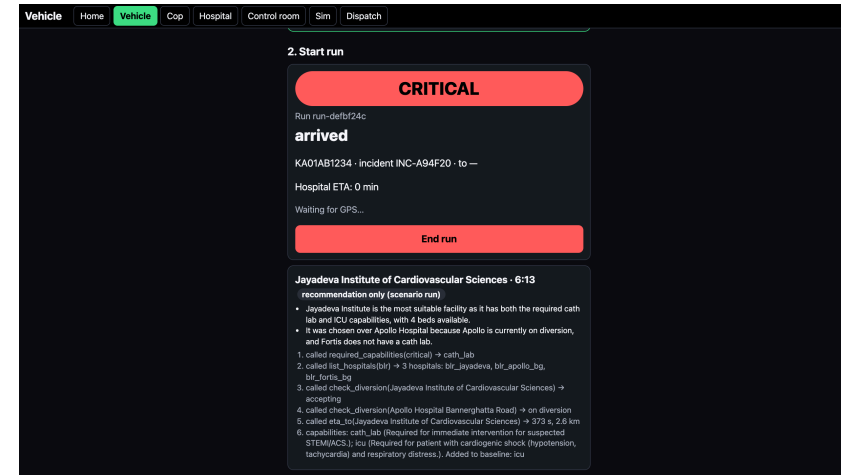
Extraction runs on `gemini-3.1-flash-lite` with `gemini-3-flash-preview` as fallback; the brief, report, sequencing sentence and agent text use `gemini-3-flash-preview`. Synthetic patients only; a clinician confirms every extracted value.

The agent: hospital routing on Agent Development Kit

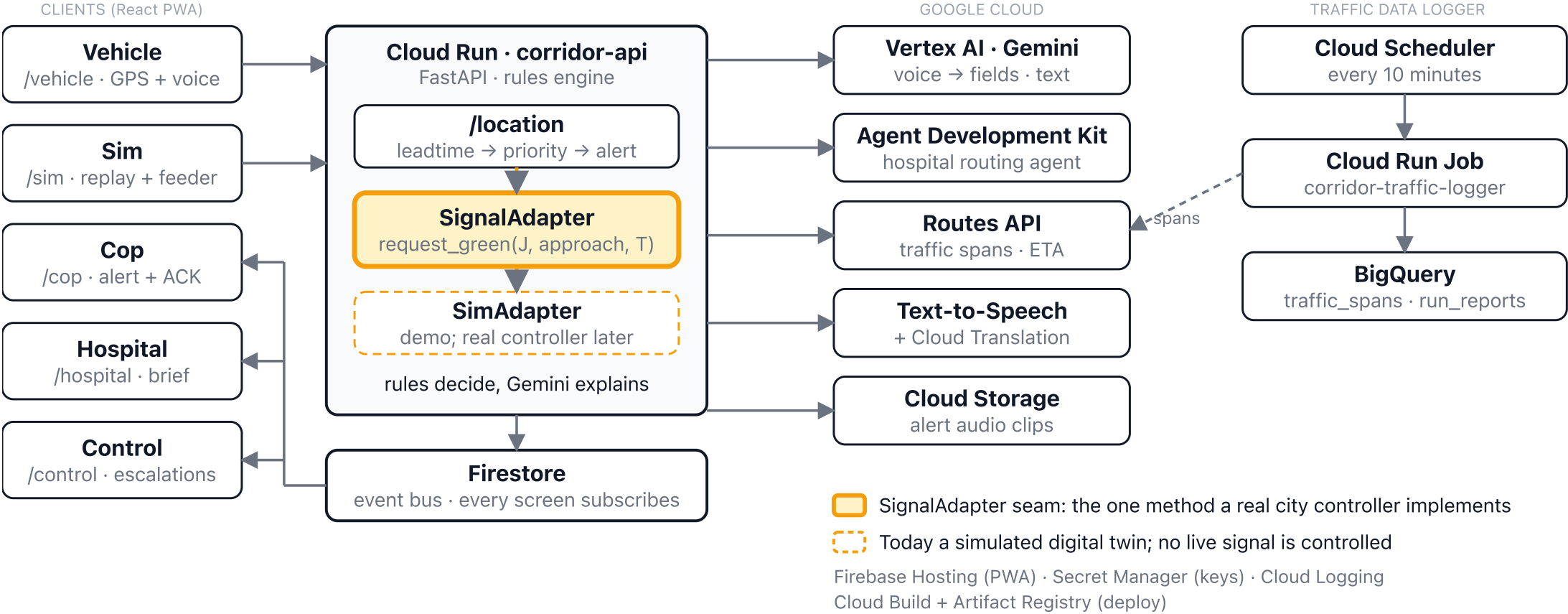
```
required_capabilities(critical) -> cath_lab  
list_hospitals(blur) -> 3 hospitals  
check_diversion(Jayadeva) -> accepting  
check_diversion(Apollo) -> on diversion  
eta_to(Jayadeva) -> 373 s, 2.6 km  
Added to baseline: icu
```

Destination Jayadeva, confidence 1.0. Alternatives:
Apollo (diversion), Fortis (no cath lab).

An ADK agent on Vertex AI picks the destination after the crew confirms the tier, with up to two rejected alternatives and a confidence. `required_capabilities` is a keyword baseline for validation. A code guard re-checks the choice; any failure or 20 s timeout falls back to the nearest eligible hospital.



Architecture



Cost by design: ~3 traffic-aware Routes calls per vehicle-minute; AI side effects run off the request path. Wired on main only. BigQuery ML: pipeline in place, retrained before submission on N rows ([fill at freeze]); not a deployed forecast.

Real data: the traffic logger

BUILT

- Cloud Scheduler → Cloud Run Job during Bengaluru peak hours (08–11, 17–21 IST), running since 6 Oct
- Live jam and slow spans per approach → BigQuery `traffic_spans`, two corridors
- Same `jam_metres` as the live engine

PROOF OF PIPELINE, NOT A FORECAST

- Rows so far: N ([fill at freeze]). The first 216 rows were midnight with zero queues
- BigQuery ML model: a pipeline proof until peak rows accumulate; retrained before submission, not a deployed forecast
- First step to a learning controller; clearance rate stays constant until observed

Run reports (the with-vs-without report card per run) also land in BigQuery `run_reports`.

Sustainability

0.76 L/h

idle burn, small petrol car (low end of 0.2–0.5 gal/h)

2.35 kg

CO₂ per litre of petrol (diesel 2.69 kg/L)

0.6 kg

CO₂ idled by one 20-vehicle queue in 60 s (illustration)

Preemption also idles the cross traffic, so the claim is the **net** effect. We size the green to the queue, so cross-traffic idling is bounded by `clear_s`.

`net_fuel_L = (ambulance-lane idling avoided - cross-traffic idling added) × idle_L_per_s`

SDG 3.8 emergency care access (universal health coverage) · **SDG 11.2** sustainable transport.

Idle burn: Argonne National Laboratory (Gaines, Rask, Keller), US DOE, 2014. CO₂: US EPA, GHG Equivalencies Calculator, 8,887 g/gal petrol and 10,180 g/gal diesel, 2024. The per-queue arithmetic (20 vehicles, 60 s) is an illustration, not a measurement or a net saving.

Deployment path, and honesty

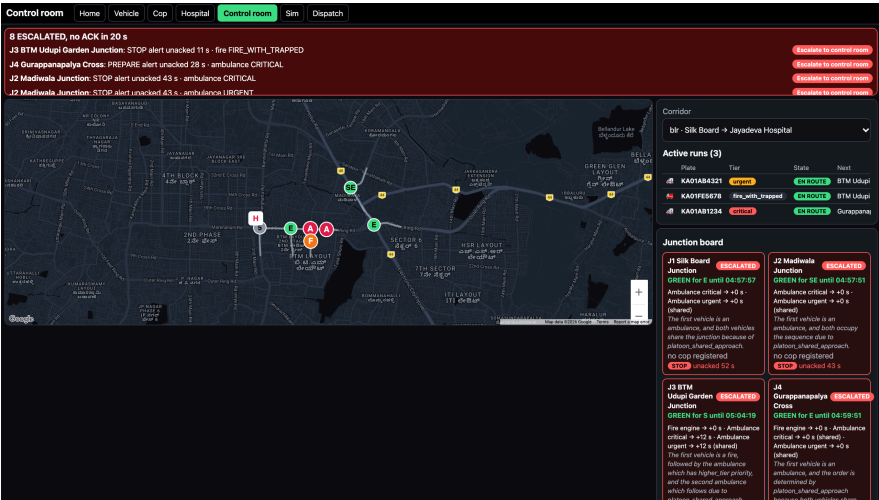
Phase	What
Today	cop alerts, no hardware, a phone
Phase 2	real controller behind <code>SignalAdapter</code>
Phase 3	learning controller on junction data

STATED PLAINLY

- Traffic signals are simulated behind an adapter; the demo scenario uses hand-authored traffic spans
- Patients are synthetic; no user accounts; device-scoped tokens protect vehicle, cop and run actions; demo endpoints are rate-limited but reachable
- Minutes saved is a simulation estimate

Global by configuration

	blr	hyd
Corridor	Silk Board → Jayadeva	Hyderabad
Alert language	English, Kannada	English, Telugu
Switch	?corridor=blr	?corridor=hyd



A corridor is one JSON file: junctions, approaches, signal cycle, language. No code change.

Try it

Live

green-corridor-2026.web.app

Repo

github.com/Nandish3010/ideal-disco

API health: corridor-api-919512130399.asia-south1.run.app/health · Demo video: [\[link, 3 minutes\]](#)

Ask: feedback from traffic police and 108 operators, and a pilot junction.